

CMP263 - Machine Learning Paper

Japanese House Prices Prediction

Juliano Almeida Machado¹

¹Instituto de Informática – Universidade Federal do Rio Grande do Sul (UFRGS)
Caixa Postal 15.064 – 91.501-970 – Porto Alegre – RS – Brazil

Abstract. *This paper studies residential real estate price prediction in Japan using transaction-level data from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT). The problem is formulated as a supervised regression task in which prices are estimated from spatial, temporal, accessibility, and physical attributes. The empirical workflow combines chronological validation, leakage-aware preprocessing, model comparison, and interpretability analysis across the main residential property types. The study compares tree-based ensembles, regularized linear regression, and support vector regression. Gradient boosting outperforms the baselines. The predictions are strongest for pre-owned condominiums and weakest for residential land, where location matters more than physical size.*

Resumo. *Este trabalho investiga a predição de preços de imóveis residenciais no Japão a partir de dados transacionais do Ministry of Land, Infrastructure, Transport and Tourism (MLIT). O problema é formulado como uma tarefa de regressão supervisionada, na qual os preços são estimados a partir de atributos espaciais, temporais, físicos e de acessibilidade. O fluxo empírico combina validação cronológica, pré-processamento voltado à prevenção de vazamento de informação, comparação de modelos e análise interpretável para os principais tipos de imóveis residenciais. O estudo compara ensembles de árvores, regressão linear regularizada e regressão por vetores de suporte. O gradient boosting supera as baselines. As previsões são mais precisas em condomínios usados e mais fracas em terrenos residenciais, nos quais a localização pesa mais que o tamanho físico.*

1. Introduction

The Japanese housing market presents an anomaly within global real estate economics. In most developed nations, residential property functions as an appreciating asset and a means for generational wealth accumulation. In contrast, housing in Japan behaves as a depreciating consumer durable. Within 20 to 30 years of construction, the structural value of a typical Japanese home depreciates to virtually zero. This structural depreciation creates a stark institutional decoupling between land value and building value [Yoshida 2016]. While urban land, especially within the Tokyo and Osaka metropolitan areas, retains substantial value and responds to conventional market forces, the structures erected upon that land are treated as temporary assets. According to data from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT), the secondary housing market accounts for only approximately 15% of total housing transactions in Japan, compared to over 80% to 90% in the US and UK [Ministry of Land, Infrastructure, Transport and Tourism 2019].

This environment traces its origins back to the post-World War II reconstruction era. Facing severe housing shortages and rapid urbanization, the Japanese government prioritized speed and quantity over long-term structural durability [Hirayama and Ronald 2006]. The resulting regulatory framework and market incentives codified a system of high-turnover construction, which was later exacerbated by the asset price bubble of the late 1980s. Japan's geographic vulnerability to seismic activity also dictates its building philosophies. The Building Standard Act of Japan has undergone continuous revisions, with the most critical shift occurring in 1981 with the introduction of the *Shin-Taishin* New Anti-Seismic Design Code). Homes constructed prior to 1981 are fundamentally viewed as structurally unsafe and frequently fail to qualify for standard bank financing, as they're heavily penalized under modern insurance structures. Subsequent amendments following the 1995 Hanshin-Awaji (Kobe) earthquake and the 2011 Tohoku earthquake have further accelerated the technological obsolescence of older properties. Consequently, buyers often perceive purchasing a pre-owned home not just as an economic compromise, but as a direct compromise on structural safety.

Complementing these rigid safety mandates is a pervasive cultural preference for new builds. A newly built home carries a significant socio-cultural premium in Japan, symbolizing clean slates, modernization, and social status. This preference is structurally reinforced by financial institutions, whereas Japanese banks routinely structure residential mortgages around the statutory useful life of the building. For a standard timber-frame home, this period is legally defined as 22 years. Once a structure surpasses this threshold, securing a mortgage for its resale becomes exceedingly difficult, effectively freezing the secondary market.

It is also important to note that one of the most pressing modern challenge facing the Japanese housing market is the nation's demographic trajectory. Japan is characterized by a rapid aging society combined with a beneath replacement fertility rate, a critical example of most modern post-industrial societies. The population has been in a continuous state of decline since its peak in the late 2000s. This population contraction has triggered the abandoned house (*akiya*) phenomenon. Millions of homes across rural, suburban, and even peripheral urban areas stand completely vacant, given the tendency of inheriting families to frequently abandon the physical structures. This has created a massive inventory of valueless, deteriorating properties that distort local real estate valuations and create severe municipal management challenges.

The Japanese housing market presents an invaluable case study for urban economists and sociologists alike. It challenges the universal assumption that residential property is an inherently safe, appreciating asset class. As other post-industrial nations begin to face similar demographic plateaus and climate-induced regulatory pressures, understanding the mechanisms of Japan's "scrap-and-build" housing model becomes essential.

2. Research Problem and Objectives

This work investigates the extent to which transaction-level information can explain and predict residential real estate prices in Japan. Using data collected by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) between 2005 and 2019 and made available through Kaggle [Nishio 2019], the study frames Japanese housing valuation as a supervised regression problem. The central task is to estimate property transaction

prices from observable attributes such as location, land characteristics, building age, and temporal market conditions.

The problem is relevant because Japanese real estate prices are shaped by a combination of economic, demographic and geographic factors that differ from those observed in many other developed housing markets. In this context, predictive modeling is not only a technical exercise, but also a way to examine which variables carry the strongest explanatory power in a market characterized by structural depreciation, regional imbalance, and changing demand. Accordingly, this paper aims to develop, evaluate, and interpret Machine Learning regression models capable of predicting housing prices while providing insight into the main determinants of valuation in the Japanese residential market.

This study followed a reproducible empirical workflow that combined data cleaning, feature preparation, model comparison, and interpretability analysis. Several regression algorithms with distinct inductive biases, including Gradient Boosting ElasticNet regression, and Support Vector Regression, were trained and evaluated under the same temporal validation strategy. Rather than estimating a single pooled model for all observations, the empirical design separated the task into independent studies for the main residential property types. The models were trained and evaluated under the same temporal validation strategy within each study. Their predictive performance was compared using standard regression metrics, while feature importance analysis was used to relate model behavior back to the economic and spatial determinants of Japanese housing prices.

3. Exploratory Data Analysis

The exploratory analysis began with an inspection of the structure, completeness, and temporal distribution of the MLIT transaction dataset. Each record describes a real estate transaction in Japan, with *TradePrice* defined as the target variable to be predicted. The explanatory variables combine geographic identifiers, such as prefecture, municipality, district, and nearest station, with accessibility measures, including the minimum and maximum travel time to the closest station.

The dataset also includes physical property attributes, such as area, land shape, frontage, floor plan, building year, structure, total floor area, and renovation status, as well as urban-planning information, including zoning, coverage ratio, floor-area ratio, road direction, and road classification. Temporal variables, represented by transaction year, quarter, and period, were used to examine the distribution of observations over time and to support the chronological modeling strategy adopted later in the pipeline. This initial analysis also guided the treatment of missing values, redundant variables, and features that could leak information from the target.

The correlation analysis in Figure 1 highlights the strongest linear relationships between *TradePrice* and the available numerical attributes. Property size, represented by *Area* and *TotalFloorArea*, has a clear positive association with transaction price, confirming that larger properties tend to command higher market values. Road frontage and other access-related variables are also related to price, suggesting that physical accessibility and road exposure contribute to property valuation. One less intuitive result is the weak negative correlation between *BuildingYear* and *TradePrice*, since newer buildings are usually expected to be more valuable. This pattern may reflect the fact that many older properties in the dataset are larger, renovated, or located in more established and

expensive urban areas. The following plots examine these hypotheses by looking at the temporal coverage of the data, the relationship between building age and property area, and the renovation profile of older buildings.

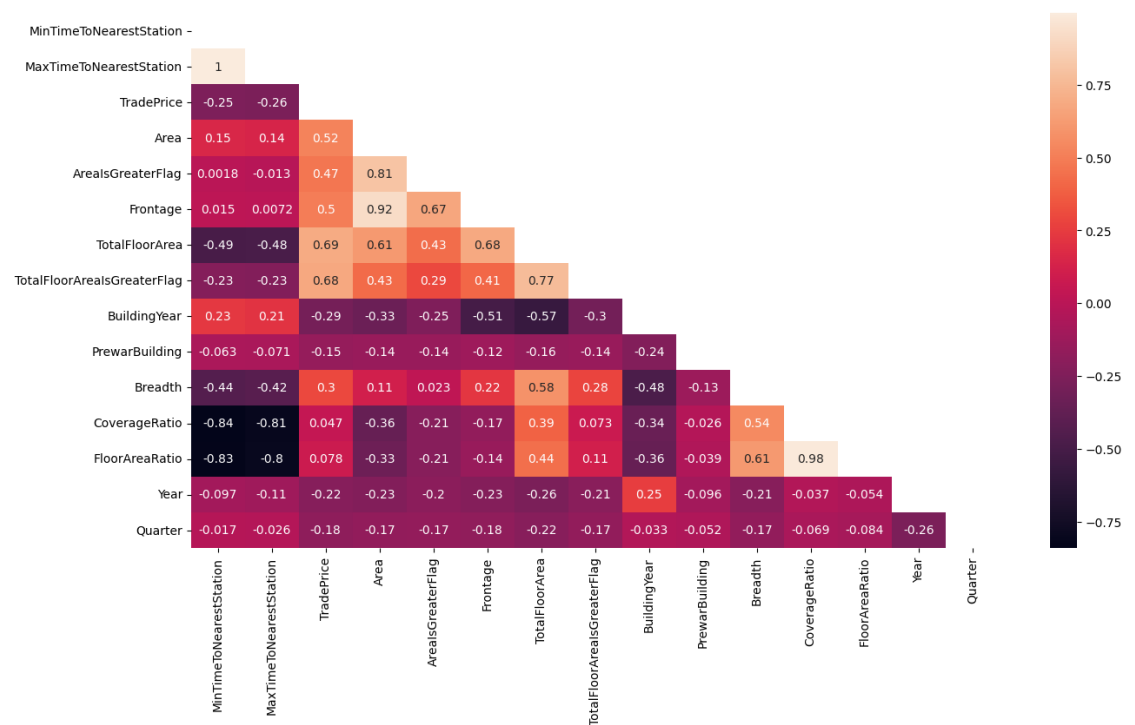


Figure 1. Correlation matrix for the numerical variables

Figure 2 summarizes the temporal structure of the dataset. The transaction-year distribution indicates how observations are spread across the market period covered by the data, while the construction-year distribution shows the age profile of the traded building stock. This is important because the project uses a chronological train-validation-test split, and because building age may capture both depreciation and location effects.

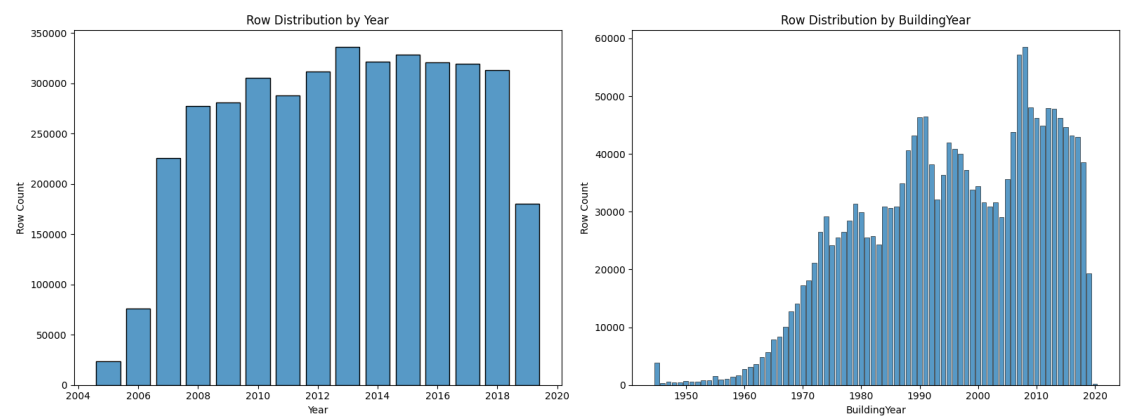


Figure 2. Temporal distributions in the dataset: transaction years and construction years of the traded properties.

Figure 3 suggests that older buildings tend to have larger floor areas. This helps

explain why the direct correlation between construction year and price can be weak, since building age is not isolated from other value drivers, especially size and location. In this dataset, some older properties may remain expensive because they offer more usable space or are concentrated in mature urban districts.

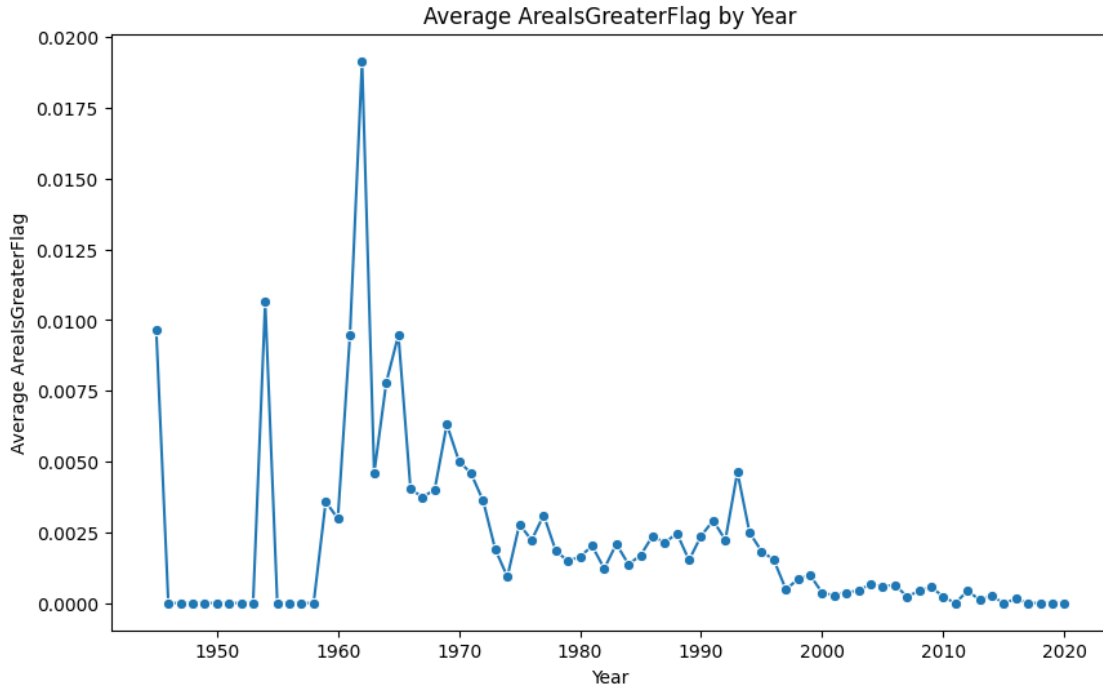


Figure 3. Relationship between building age and property area, showing that older properties often correspond to larger buildings in the observed transactions.

4. Methodology

4.1. Data Cleaning and Feature Preparation

Data cleaning was performed only after the chronological train-validation-test split, so that all preprocessing parameters were estimated from past observations and then applied to later periods. This design reduced the risk of information leakage from future transactions into the training process. SQLite was used as a lightweight database layer to organize the raw files and enable efficient access to the large dataset directly within the code. The first step removed variables that were either not suitable for modeling or directly derived from the target. The textual *Remarks* field was discarded because of its unstructured nature and high missingness, same as for *Renovation*, while *UnitPrice* and *PricePerTsubo* were removed because they encode transformations of *TradePrice* and would therefore leak information about the response variable.

Missing values were then treated through a combination of domain rules and train-fitted statistical imputations. For property types in which some attributes were conceptually absent, such as land-only, condominium, or rural transactions, variables including floor plan, structure, road classification, city planning, frontage, breadth, and nearest station were filled as not applicable or with structurally meaningful values. In particular,

frontage and road breadth were set to zero for condominiums, where these measurements do not translate to the category. The original station-distance field was also reconstructed from the minimum and maximum time-to-station variables, producing a consistent numerical accessibility measure. Remaining numerical gaps, including frontage, breadth, coverage ratio, and floor-area ratio, were imputed with statistics learned exclusively from the training set, primarily district-level aggregates with global medians as fallbacks. Remaining missing values in structure, road classification, and city planning were kept as an explicit unknown category before categorical encoding. Because the MLIT transaction records do not include geographic coordinates, latitude and longitude were obtained from an auxiliary source, the Geolonia Japanese address dataset. Each transaction was matched to a coordinate pair by combining Prefecture, Municipality, and District, yielding district-level *Latitude* and *Longitude* features that were merged into the modeling tables. Missing coordinates after this merge were handled with municipality-level medians, again fitted only on the training partition and then applied consistently to the validation and test sets.

Additional feature preparation was applied after the initial cleaning stage. The original station-distance representation was converted into a numerical *TimeToNearestStation* variable using the midpoint between the minimum and maximum reported travel times, while the transaction quarter was encoded with sine and cosine transformations to preserve its cyclical structure. The target variable was transformed with $\log(1 + x)$ because housing prices are strongly right-skewed, with a small number of very expensive transactions that can dominate loss functions based on absolute or squared errors. Modeling the logarithm of price compresses these extreme values and makes the learning problem closer to predicting relative price differences rather than only fitting the largest transactions. The models were therefore trained and produced predictions on the transformed scale. Before reporting evaluation metrics, both predictions and observed values were converted back to Yen using the inverse transformation. The final feature set combined numerical attributes (*TimeToNearestStation*, *Area*, *BuildingYear*, *CoverageRatio*, *FloorAreaRatio*, *Year*, cyclical quarter encodings, *Latitude*, *Longitude*, *Frontage*, and *Breadth*) with one-hot encoded categorical attributes for *Structure*, road *Classification*, and *CityPlanning*. Finally, numerical features were normalized with train-fitted min-max scalers, and feature names were standardized to ensure compatibility with the different modeling libraries.

4.2. Property-Type Study Design

The final modeling pipeline was organized as a set of independent property-type studies. Three residential market segments were considered: *Residential Land (Land and Building)*, *Residential Land (Land Only)*, and *Pre-owned Condominiums, etc.*. Agricultural and forest land observations were excluded from this stage because their price formation mechanisms differ substantially from residential transactions and would introduce a distinct economic problem into the same regression task.

For each study, the dataset was first filtered to the corresponding property type and then passed through the complete workflow of chronological splitting, cleaning, feature engineering, scaling, hyperparameter optimization, training, evaluation, and interpretability analysis. This avoids forcing structurally heterogeneous markets into a single regression function. It also prevents the property-type indicator from acting as a coarse proxy for different valuation regimes, as we've noted in the preliminary tests, where the *Type*

would act was one the of the main indicators of value. The same experimental setup was applied to all studies, while parameter files, diagnostic plots, input snapshots, and metric records were stored separately or tagged by study name to preserve traceability.

4.3. Models and Evaluation

The study compared four regression models with different assumptions and learning architectures. *XGBoost* [Chen and Guestrin 2016] was used as a gradient boosting model that builds weak decision trees sequentially, using previous errors to construct a stronger ensemble, while *LightGBM* [Ke et al. 2017] provided a more computationally efficient boosting approach based on leaf-wise tree growth. As a linear baseline, *ElasticNet* [Zou and Hastie 2005] combined L1 and L2 regularization to balance feature selection and coefficient shrinkage. Finally, *Support Vector Regression* [Smola and Schölkopf 2004] estimated a regression function within an error-tolerant margin, allowing nonlinear patterns to be modeled through kernel transformations.

Model performance was evaluated using complementary regression metrics on both the transformed and original price scales. In log space, Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) were used to compare models on the same scale used during training. After reversing the logarithmic target transformation, RMSE and Mean Absolute Error (MAE) measured absolute deviations in Yen, while original-scale R^2 summarized the proportion of price variance explained by each model.

Reproducibility was addressed by centralizing the experimental configuration, fixing random seeds for Python and NumPy, and applying the same chronological split, preprocessing rules, scaling, and model evaluation procedure across all algorithms and property-type studies. The pipeline also generated a run identifier and stored metadata, model input splits, and training, validation, and test metrics, allowing each experiment to be traced back to the study, data partitions, and feature set used during training.

Interpretability was incorporated after model selection (*post hoc*) in order to connect predictive performance with substantive housing-market explanations. For the best-performing model, global feature importance was used to identify the variables that contributed most strongly to prediction across the test set.

5. Results

Across all three markets, the gradient boosting models (GBMs) consistently outperformed the linear and kernel baselines. *ElasticNet* and *Support Vector Regression* achieved substantially higher error on both the log-transformed and original Yen scales and struggled to reproduce the wide, right-skewed distribution of transaction prices. On the test set, their predictions failed to track the observed price range, either compressing values near the lower end of the distribution or producing unstable extrapolations for higher-priced transactions. Figure 4 illustrates this pattern through an actual-versus-predicted scatter plot for the baseline models on the held-out test period in the *Residential Land (Land and Building)* study, a similar pattern emerges for the *Residential Land (Land Only)*.

Table 1 summarizes the test-set performance across the different models and property-type studies. The regression errors are highest in the residential land segments, where no model managed to generalize a highly robust solution. Performance on *Residential Land (Land Only)* is slightly worse than on *Residential Land (Land and Building)*,

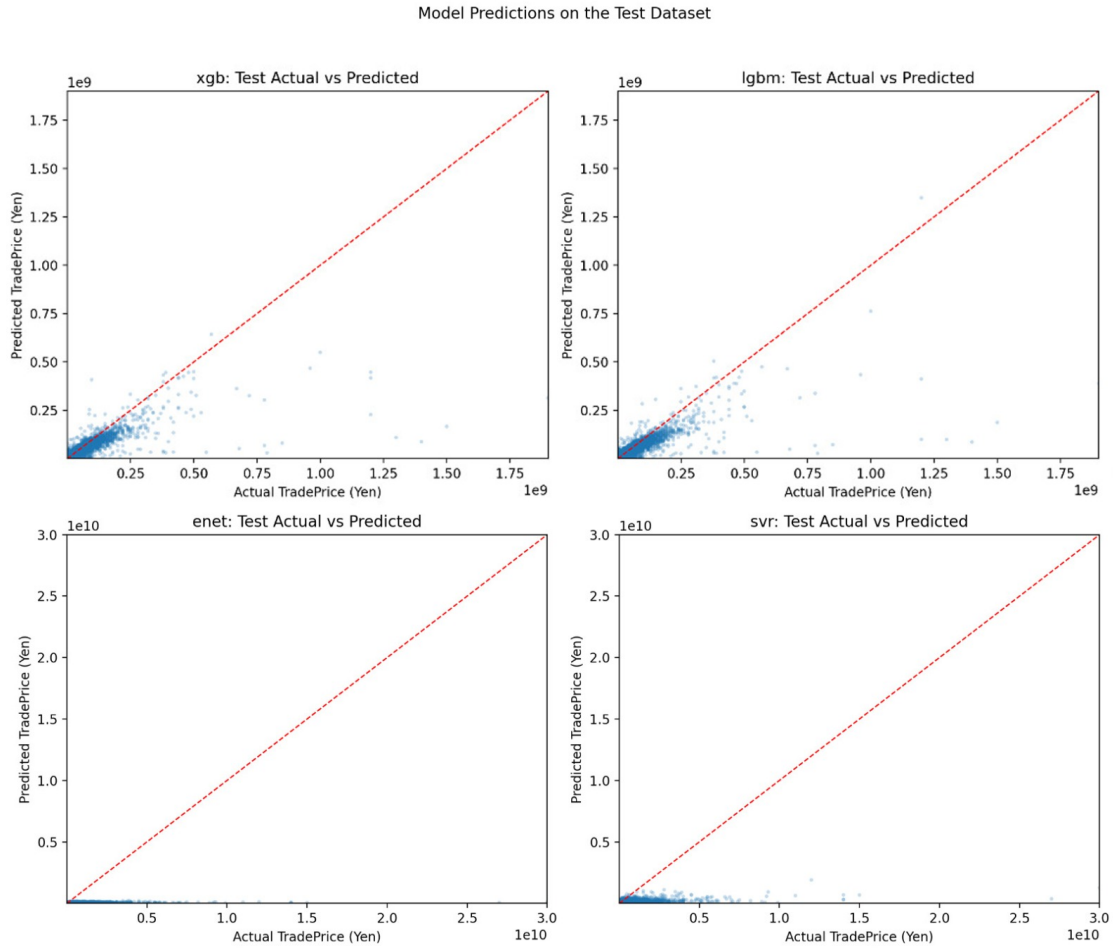


Figure 4. Test-set actual versus predicted trade prices in the *Residential Land (Land and Building)* study

though both remain far behind the condominium study. Conversely, the models trained on *Pre-owned Condominiums* yield significantly better metrics, demonstrating a genuine ability to make accurate, reliable predictions.

Comparing the feature importance profiles between these distinct market segments highlights why the condominium models achieve superior predictive power. As illustrated in Figure 5, valuation dynamics differ fundamentally between property types. For *Pre-owned Condominiums* (typically Japanese apartments), a single structural attribute, *Area*, dominates the model’s decision-making process, capturing the core variance of the problem and enabling clean generalization. That is, the asset class has a more predictable, standardized pricing mechanism.

In contrast, for *R. Land (Land and Building)*, physical dimensions alone are insufficient. Instead, spatial coordinates (*Latitude* and *Longitude*) emerge as primary drivers, confirming that real estate valuation in this segment is very sensitive to location. Since modeling localized geographical premiums and complex neighborhood dynamics are inherently more difficult than scaling prices by floor area, the models struggle to find an optimal global solution, resulting in the higher error metrics observed in Table 1.

Tabela 1. Test-set regression metrics by property-type study and model. Log-space metrics correspond to the $\log(1 + x)$ target; Yen-scale metrics are computed after inverse transformation. RMSE and MAE in Yen are reported in millions.

Study	Model	$\text{RMSE}_{\log}$	$R^2_{\log}$	RMSE (M¥)	MAE (M¥)	R^2 (Yen)
<i>R. Land (Land and Building)</i>	<i>XGBoost</i>	0.554	0.757	110.8	14.6	0.538
	<i>LightGBM</i>	0.555	0.756	106.3	14.4	0.575
	<i>ElasticNet</i>	0.790	0.575	149.7	23.3	0.157
	<i>Support Vector Regression</i>	0.804	0.582	150.2	23.0	0.151
<i>R. Land (Land Only)</i>	<i>XGBoost</i>	0.690	0.722	80.1	10.6	0.512
	<i>LightGBM</i>	0.692	0.720	83.2	10.7	0.474
	<i>ElasticNet</i>	0.969	0.435	108.3	17.2	0.107
	<i>Support Vector Regression</i>	0.970	0.434	107.8	17.2	0.116
<i>Pre-owned Condominiums</i>	<i>XGBoost</i>	0.363	0.800	13.6	5.4	0.673
	<i>LightGBM</i>	0.356	0.808	12.9	5.1	0.703
	<i>ElasticNet</i>	0.591	0.471	104.3	14.2	0.563
	<i>Support Vector Regression</i>	0.590	0.472	103.8	14.1	0.561

6. Conclusion

This study has investigated the task of predicting residential real estate transaction prices in Japan, a market characterized by a unique decoupling of structural and land values driven by strict anti-seismic mandates, cultural fresh-build preferences, and stark demographic contractions. Our empirical findings demonstrate that real estate valuation in this environment is an inherently complex problem. The predictive difficulty is heavily stratified across property types. That is, while condominium valuations scale predictably with physical size, standalone residential properties depend on localized geographical premiums that are notoriously difficult to capture using global estimators.

The primary insight of this work is that confronting this inherent difficulty requires a strategic decomposition of the market. By segmenting the macro-level dataset into isolated, homogeneous property-type studies, we prevented structural differences from muddying the models' inductive biases. This design paved a clear path toward localized optimization, yielding highly viable predictions in the condominium sector and establishing baselines for land transactions.

To bridge the remaining performance gaps in the more volatile residential land segments, future research should transition from global spatial coordinates to richer localized indicators. Integrating localized features, such as moving historical neighborhood averages, district-level socioeconomic indices, and granular accessibility metrics, will allow the models to better map complex spatial dynamics. Ultimately, this combination of targeted data segmentation and localized feature engineering offers a reproducible framework for executing predictive modeling inside highly non-traditional asset markets.

Code availability. The source code for this study is open and available at <https://github.com/jualmac/Japan-Real-Estate-Prices>.

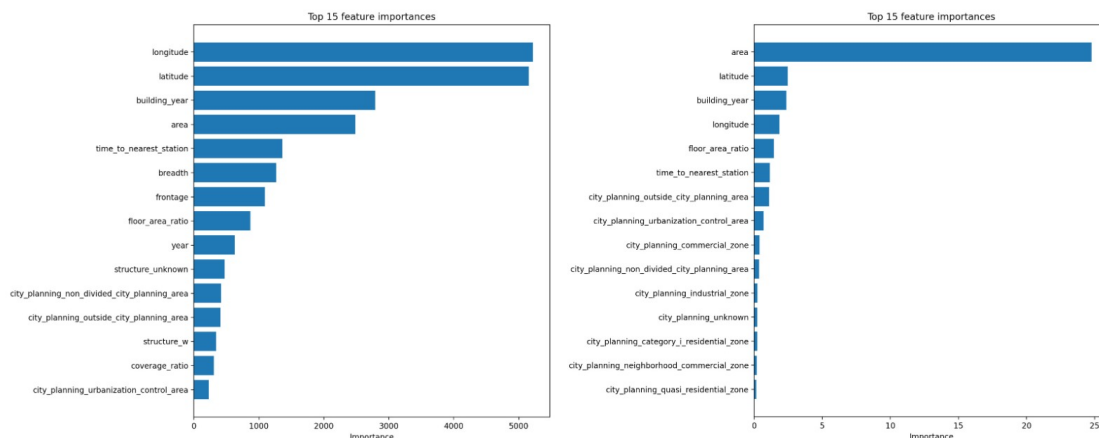


Figura 5. Comparison of the top 15 feature importances for Residential Land (Land and Building) versus Pre-owned Condominiums.

Referências

- Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, pages 785–794, New York, NY, USA. ACM.
- Hirayama, Y. and Ronald, R. (2006). *Housing and Social Transition in Japan*. Routledge.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). Lightgbm: a highly efficient gradient boosting decision tree. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17, page 3149–3157, Red Hook, NY, USA. Curran Associates Inc.
- Ministry of Land, Infrastructure, Transport and Tourism (2019). Revitalization of the market for existing housing (kizon jutaku ryutsu shijo no kasseika). <https://www.mlit.go.jp/seisakutokatsu/hyouka/content/001313273.pdf>.
- Nishio, S. (2019). Japan real estate transaction prices. <https://www.kaggle.com/datasets/nishiodens/japan-real-estate-transaction-prices>. Data sourced originally from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) of Japan with this Gist <https://gist.github.com/nishiodens/d66a52cb26a4fe0851ceb353c75ecd7>.
- Smola, A. J. and Schölkopf, B. (2004). A tutorial on support vector regression. *Statistics and Computing*, 14:199–222.
- Yoshida, J. (2016). Structure depreciation and returns to scale of real estate. *SSRN Electronic Journal*.
- Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 67:301–320.